
A cyclic-coloring obstruction to maximal binary DFAO reversal

A proposed solution to Davies’s nonattainment conjecture,
with two independent proofs, constructive certificates,
exhaustive small-state enumeration, and finite optimality bounds

ChatGPT

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Abstract

An n -state deterministic finite automaton with k output values admits a reversal construction with k^n states. We investigate whether this bound can be attained with only two input letters when $3 \leq k \leq n$. We give a self-contained proposed proof that it cannot. More precisely, for arbitrary transformations a, b of an n -element set and arbitrary k -valued output map τ , we prove

$$|\tau\langle a, b \rangle| \leq k^n - k! + g(k) < k^n,$$

where $g(k)$ is the maximum order of a permutation of k objects. The central construction is the graph formed by translating one collision pair under a permutation. This graph is always 3-colorable. Its proper colorings can only be reached through permutation powers, whereas three suitable output relabelings cannot lie in a single cyclic orbit.

Two independent proofs of nonattainment are given: a short qualitative one, which uses no counting and no Landau function, and the quantitative one that yields the displayed additive deficit. We also give a polynomial-time algorithm producing an explicit missing coloring together with two independently checkable certificate formats, extend the obstruction to an abelian group of permutation letters and one singular letter, and derive two sharper upper bounds from chromatic polynomials: an instance-specific one that is exact for the worked examples, and a weaker one that abstracts into a universal structural bound. A complete C++ enumeration settles all 462 066 reduced instances with $n \leq 4$, and exact arithmetic verifies agreement with Davies’s proposed binary optimum for all 372 pairs $7 \leq n \leq 30$, $3 \leq k < n$. This finite optimality statement is distinct from the unrestricted theorem. Source, verification programs, explicit automata, certificates, and numerical records accompany the article.

Scope. The primary target is the nonattainment question, not the exact maximum number of reverse states for every pair (n, k) . The latter remains undetermined by the methods of this article outside the finite range checked in section 10.

Keywords. Finite automata with output; reversal; state complexity; transformation monoids; graph coloring; cyclic actions; certificates.

Research status. The target is explicitly posed in Davies’s 2017 paper. A literature search on 20 September 2026 found no later resolution, but was not an exhaustive bibliographic review, a systematic citation-index review, a survey of theses, or correspondence with the author; no conclusion about priority should be drawn from a missing search hit. The proofs below have not been independently refereed or checked by a proof assistant. The status of the main inequality is therefore a *candidate theorem with a self-contained proposed proof*, not an independently certified result. The principal obstruction theorem does not depend on the computations.

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Reading routes. The shared core is sections 2 and 3. Two independent proofs of nonattainment then follow: the short qualitative one in section 4, which a referee can check first, and the quantitative one in section 5. Section 6 makes both constructive. Sections 9 and 10 contain the stronger numerical bounds and their finite verification, and section 11 the computational audit. Section C proves one orbit size by hand, with no program; section D lists the places where the argument could go wrong. The archive's `README.md` gives build and reproduction commands.

1 The problem and the result

A deterministic finite automaton normally gives a binary answer: accept or reject. A deterministic finite automaton with output, or DFAO, instead gives one value from a finite output alphabet. The output is attached to the final state reached after the entire input word has been read. Thus a DFAO computes a function $f : \Sigma^* \rightarrow \Delta$, not a stream of output symbols.

Reversing the direction of input means computing

$$f^R(w) = f(w^R),$$

where w^R is the word written backwards. The number of states can increase substantially. If the original machine has n states and k output values, the natural reversal construction has a potential state for each function from the original state set to the output alphabet. There are k^n such functions.

Davies explicitly asks whether this bound is attainable with two input letters and $k \geq 3$, and conjectures that it is not [1, Section 5, Problem 1, printed page 17]. That is the problem selected here. The diagonal case $k = n$ was already settled in that source; the remaining range was $3 \leq k < n$. Her second problem concerns the exact binary maximum; we treat it separately. The binary restriction is on the *input* alphabet. The output alphabet has k symbols throughout.

1.1 Algebraic formulation

Let Q and Δ be finite sets of sizes n and k . A transformation is a function $Q \rightarrow Q$. Write $T(Q)$ for the monoid of all transformations of Q , with usual function composition:

$$(u \circ v)(q) = u(v(q)).$$

If $M \subseteq T(Q)$ and $\tau : Q \rightarrow \Delta$, define

$$\tau M = \{\tau \circ t : t \in M\} \subseteq \Delta^Q.$$

The monoid $\langle a, b \rangle$ includes the identity transformation. The automata question reduces to whether

$$\tau \langle a, b \rangle = \Delta^Q \tag{1.1}$$

can hold. We prove that it cannot when $3 \leq k \leq n$.

Theorem 1.1 (Binary obstruction and quantitative gap). *Let $3 \leq k \leq n$, let $a, b : Q \rightarrow Q$, and let $\tau : Q \rightarrow \Delta$ be arbitrary. Define*

$$g(k) = \max\{\text{ord}(\sigma) : \sigma \in \text{Sym}(\Delta)\}.$$

Then

$$|\tau \langle a, b \rangle| \leq k^n - k! + g(k) < k^n. \tag{1.2}$$

Consequently, the state complexity of the reversal of any n -state binary DFAO with k output symbols is strictly less than k^n .

The function g is often called Landau's function; the standard terminology is used, for example, in [3]. Only the finite-group definition is needed, and no estimate for its growth: a cyclic subgroup cannot equal the nonabelian group $\text{Sym}(\Delta)$ when $k \geq 3$, so $g(k) < k!$.

The conclusion is deliberately stronger than necessary for minimal automata: it applies to every transformation pair and every output map, surjective or not. Accessibility, distinguishability, and the initial state enter only when translating the orbit size into exact automaton state complexity.

Corollary 1.2 (Automaton interpretation and the alphabet threshold). *If a function f is computed by an n -state DFAO with at most two input letters and $3 \leq k \leq n$ output labels, then*

$$\text{sc}(f^R) \leq k^n - k! + g(k) < k^n.$$

Moreover, for each $n \geq k \geq 3$, the least number of input letters needed by a minimal n -state DFAO to attain k^n reverse states is exactly three; see theorem 7.2 for $k = 2$.

The first assertion follows from the reversal construction of section 2. The second combines the obstruction with the three-letter construction of section 7; a one-letter machine is covered by duplicating its transformation as a second letter.

For $k = n$, the numerical bound specializes to the classical transformation-monoid bound $n^n - n! + g(n)$ of Holzer and König [2, Theorem 13]. The substantive extension here is to $k < n$: counting omitted permutations of Q does not directly survive the projection $t \mapsto \tau \circ t$. The proof instead makes the symmetric group act on the k output labels.

The proof has three cases. If both letters are permutations, they preserve the number of used output colors. If both are singular, two collision pairs already obstruct reachability. The interesting case is one permutation and one singular transformation. A single collision then generates a small chromatic obstruction under repeated rotation.

1.2 Size of the guaranteed deficit

A permutation with disjoint cycle lengths $\lambda_1, \dots, \lambda_r$ has order $\text{lcm}(\lambda_1, \dots, \lambda_r)$, and every partition of k occurs as a cycle type. Consequently

$$g(k) = \max_{\lambda_1 + \dots + \lambda_r = k} \text{lcm}(\lambda_1, \dots, \lambda_r), \quad (1.3)$$

a small exact computation over integer partitions; the bundled implementation uses no asymptotic approximation. Exact values of $g(k)$ and of the deficit $k! - g(k)$ for $3 \leq k \leq 10$ are tabulated in table 1; the machine-readable file `results/landau_gaps.json` extends the same data to $k = 12$.

For $k = 3$ the theorem gives $3^n - 3$. At $n = 3$ this is sharp, as the word certificate of section C shows by hand. At $n = 4$ the uniform bound gives 78, whereas the graph bound of section 9 gives 67 for a maximizing witness, which is also the true value. The uniform theorem therefore answers the attainment question without purporting to solve the entire optimization problem.

1.3 Provenance of this report

This article merges two independently prepared research packages, both dated 20 September 2026 and both targeting the same question.

`dfao-reversal-cyclic-coloring` proved the shared theorem by the three-case argument reproduced in section 5, packaging the relabeling bound as a subgroup $K \leq \mathbb{Z}$ of exponents with a homomorphism $\phi : K \rightarrow \text{Sym}(\Delta)$ (theorem 5.1); it supplied in addition the independent qualitative proof of section 4, the abelian extension of section 8, the universal structural bound $U(n, k)$ of section 9, the 372-pair finite optimality result of section 10, the correction to the source’s printed Table 2, the Holzer–König attribution, and the independent C++ breadth-first search.

`dfao-reversal-coloring-bound` proved the same theorem by the same three-case argument with the same collision-orbit graph, packaging the relabeling bound instead as the *setwise* stabilizer $H \leq \langle p \rangle$ of the fiber partition of a surjective coloring, with induced homomorphism $\rho : H \rightarrow \text{Sym}(\Delta)$ (theorem 5.2); it supplied in addition the sharper instance-specific chromatic bound theorem 9.1, the complete $n \leq 4$

enumeration of section 11, the standalone certificate checker with its pairwise-compatibility Chinese remainder criterion, the hand-checkable 24-word certificate of section C, and the research-status and literature-search notes retained in the archive.

The two proofs of the shared inequality are the same mathematics in different dress, and it is proved here exactly once, in section 5. Everything built on top of that core was complementary, and is kept. Where the two packages stated the status of a source or of the literature differently, the more cautious statement is the one retained.

1.4 What is and is not established

The main theorem is unrestricted in n and k within $3 \leq k \leq n$. It applies even without accessibility, minimality, or surjectivity assumptions on the supplied machine. The article also proves an extension to commuting permutation letters, and gives a deterministic algorithm for constructing one unreachable coloring.

The exact maximum number of binary reversal states is a different question. We derive an explicit universal upper bound $U(n, k)$ by classifying the collision graphs. Exact evaluation gives the same number as the published coprime-block lower bound for $7 \leq n \leq 30$ and $3 \leq k < n$. This is a finite, computer-assisted optimality result. We do not prove equality with that lower bound for all n .

The following are *not* claimed. No proof of the exact optimal binary reversal complexity for arbitrary n, k , and no resolution of Davies’s other open questions. No novelty for the orbit reduction, the ternary witness, the standard Landau function, the elementary group facts, the previously reported small maxima, the diagonal specialization $k = n$ to the classical bound $n^n - n! + g(n)$ of Holzer and König, or graph coloring and chromatic polynomials as a method in the transformation-monoid setting. No claim that the absence of a located later solution proves the problem remained open until this draft. No Lean, Isabelle, Coq, or other proof-assistant certification. No claim that random tests establish a universal theorem. No claim that all improper graph colorings are reachable in arbitrary cases. The archive’s `source_audit.md` records the search scope, the locations actually inspected in the source, and the attribution boundaries of this merged report in full. `RESEARCH_STATUS.md` and `LITERATURE_SEARCH.md` are kept verbatim from one of the two source reports and describe that report rather than this one; each carries a note saying so.

2 Why reversal is an orbit of colorings

2.1 The reversal construction

Write a DFAO as

$$\mathcal{A} = (Q, \Sigma, \delta, q_0, \Delta, \tau).$$

Each input letter x induces a transformation $t_x(q) = \delta(q, x)$. For a word $w = x_1 \cdots x_r$, the forward transformation is

$$t_w = t_{x_r} \circ \cdots \circ t_{x_1}.$$

The order here matters. The machine reads x_1 first, whereas function composition applies its rightmost factor first. The function computed by $\mathcal{A}$ is $f(w) = \tau(t_w(q_0))$, and we write $w^R = x_r \cdots x_1$ and $f^R(w) = f(w^R)$.

An automaton is *accessible* if every state is reachable from q_0 , and two states are *distinguishable* if some continuation word gives different outputs. An accessible automaton whose distinct states are pairwise distinguishable is *minimal*: two distinguishable states cannot be identified in any deterministic machine with the same continuation behaviour. Accordingly $\text{sc}(f)$ denotes the *state complexity* of f , the least number of states in a DFAO computing it.

Define a new DFAO on the set Δ^Q . Its initial state is τ , its transition on x sends a coloring c to $c \circ t_x$, and its output at c is $c(q_0)$. After reading $x_1 \cdots x_r$, its state is

$$\tau \circ t_{x_1} \circ \cdots \circ t_{x_r} = \tau \circ t_{w^R}.$$

Its output is therefore $\tau(t_{w^R}(q_0)) = f(w^R)$, as required.

If M is the transition monoid generated by the original letters, the reachable state set of this reversal construction is exactly τM . It follows immediately that

$$\text{sc}(f^R) \leq |\tau M| \leq k^n.$$

Proposition 2.1 (Reachable colorings are distinguishable). *If every state of the original DFAO is reachable from q_0 , then*

$$\text{sc}(f^R) = |\tau M|.$$

Proof. Let $c, d \in \tau M$ be distinct. Choose q with $c(q) \neq d(q)$. By accessibility, some word u satisfies $t_u(q_0) = q$. Read u^R in the reversal construction starting from c or from d . The resulting outputs are respectively

$$c(t_u(q_0)) = c(q), \quad d(t_u(q_0)) = d(q).$$

They differ, so c and d cannot be identified by minimization. Every reachable state is therefore necessary. $\square$

This orbit characterization is also established by Davies [1, Proposition 4]; the proof is included to fix all conventions and to make the later argument independent of a black-box automata theorem.

2.2 Rank, fibers, and output relabeling

The rank of a coloring $c : Q \rightarrow \Delta$ is $|c(Q)|$. Precomposition by a permutation preserves this rank and, more strongly, preserves the size of every labeled color class. Precomposition by a singular transformation can merge distinctions: if $t(x) = t(y)$, then

$$(c \circ t)(x) = (c \circ t)(y)$$

for every coloring c .

There is a second action, on the opposite side. An output permutation $\sigma \in \text{Sym}(\Delta)$ acts by relabeling colors:

$$c \mapsto \sigma \circ c.$$

The two actions commute, in the precise sense

$$(\sigma \circ c) \circ t = \sigma \circ (c \circ t). \tag{2.1}$$

This elementary associativity identity is the algebraic core of the proof.

If c is surjective, its output-relabeling orbit

$$\mathcal{R}(c) = \{\sigma \circ c : \sigma \in \text{Sym}(\Delta)\}$$

has exactly $k!$ elements. Indeed, $\sigma \circ c = \rho \circ c$ implies $\sigma = \rho$ by evaluating at a preimage of each output value.

3 The collision-orbit graph

3.1 Definition and low chromatic number

Suppose p is a permutation of Q and s is a singular transformation. Choose one collision pair

$$x \neq y, \quad s(x) = s(y).$$

Define a simple undirected graph $G = G(p; x, y)$ with vertex set Q and edge set

$$E(G) = \{\{p^j(x), p^j(y)\} : j \in \mathbb{Z}\}. \quad (3.1)$$

Repeated edges are retained only once. There are no loops because p is injective. The graph is invariant under both p and p^{-1} .

Lemma 3.1. *Every graph $G(p; x, y)$ is 3-colorable.*

Proof. If x and y lie in different cycles of p , every edge runs between those two cycles. They form the two parts of a bipartite graph, and all other vertices are isolated.

Suppose instead that x and y lie on the same cycle of length m . Identify that cycle with $\mathbb{Z}/m\mathbb{Z}$, with $p(i) = i + 1$, and write the pair as $\{0, h\}$, where $h \not\equiv 0 \pmod{m}$. Its translates are $\{i, i + h\}$. Each vertex has at most two neighbors, $i + h$ and $i - h$. Hence the nontrivial components are cycles or single edges. Such a graph is 3-colorable. Isolated vertices cause no restriction. $\square$

Lemma 3.2 (Refinement to a surjective coloring). *If a graph on n vertices is r -colorable and $r \leq k \leq n$, then it has a proper coloring using exactly k colors.*

Proof. Start with a proper coloring using $j \leq r$ nonempty color classes. If $j < k$, then $j < n$, so some class contains at least two vertices. Give one of those vertices a fresh color. Splitting a color class cannot create a monochromatic edge. Repeating the operation reaches exactly k nonempty classes. $\square$

In particular, for every $3 \leq k \leq n$ the graph G has a proper surjective k -coloring. Every output relabeling of that coloring is still proper. The same refinement applies to the two-edge graph used in the case of two singular generators: a simple graph with at most two edges contains no cycle, hence no odd cycle, and is bipartite.

3.2 A singular letter cannot reach a proper coloring

Lemma 3.3 (Rightmost singular factor). *Every coloring in $\tau\langle p, s \rangle$ obtained from a product containing s is improper on $G(p; x, y)$. Consequently,*

$$\tau\langle p, s \rangle \cap \text{Col}_k(G) \subseteq \{\tau \circ p^j : j \in \mathbb{Z}\}, \quad (3.2)$$

where $\text{Col}_k(G)$ is the set of all proper k -colorings, not necessarily surjective.

Proof. Take a product containing s . Locate its rightmost occurrence of s in usual composition notation. All factors to its right are powers of p , so the product has the form

$$t = u \circ s \circ p^j$$

for some $u \in \langle p, s \rangle$ and $j \geq 0$. Put $x' = p^{-j}(x)$ and $y' = p^{-j}(y)$. Then

$$t(x') = u(s(x)) = u(s(y)) = t(y').$$

The pair $\{x', y'\}$ is an edge of G . Thus $(\tau \circ t)(x') = (\tau \circ t)(y')$, which makes the coloring improper. Products with no s are powers of p , proving the inclusion. $\square$

A proper coloring remains proper under precomposition by p . Therefore, if τ itself is proper, the reachable proper colorings form its whole cyclic orbit; if τ is improper, that set is empty. We will usually need only the weaker inclusion (3.2).

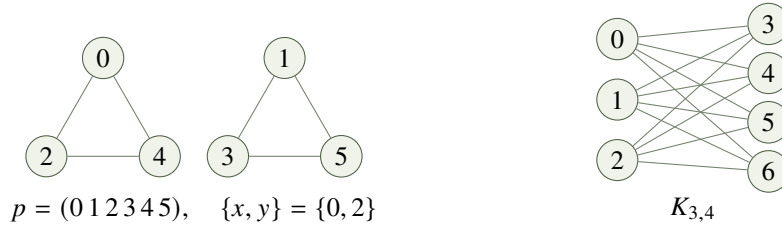


Figure 1: The two basic mechanisms: a same-cycle pair produces disjoint cycles; a pair from different permutation cycles produces a bipartite graph. These diagrams illustrate the construction, not an additional assumption.

4 First proof of nonattainment: a short qualitative route

Before counting missing states, we prove that at least one must be missing. The useful ingredient is a small obstruction to fitting output symmetries into a cyclic input action.

First of two independent proofs. This section uses no counting, no Landau function, and no refinement beyond one surjective proper base coloring. What it buys: it is the shorter statement a referee can check first, it is logically independent of the counting argument of section 5, and it is what makes the constructive algorithm of section 6 cheap, needing only three candidate colorings instead of the six relabelings that the counting route suggests.

Lemma 4.1 (Three relabelings cannot share a cyclic orbit). *Let $c : Q \rightarrow \Delta$ be surjective with $k \geq 3$, and let p be a permutation of Q . Choose three distinct output symbols, written 0, 1, 2, and let $\sigma = (0\ 1)$ and $\rho = (1\ 2)$ fix all other symbols. No orbit of precomposition by p contains all three colorings*

$$c, \quad \sigma \circ c, \quad \rho \circ c.$$

Proof. Suppose one orbit contains all three. Taking c as its base point, there exist integers r, t such that

$$\sigma \circ c = c \circ p^r, \quad \rho \circ c = c \circ p^t.$$

Using (2.1),

$$\begin{aligned} \sigma \circ \rho \circ c &= \sigma \circ c \circ p^t = c \circ p^{r+t}, \\ \rho \circ \sigma \circ c &= \rho \circ c \circ p^r = c \circ p^{t+r}. \end{aligned}$$

The two expressions are equal. Surjectivity of c then forces $\sigma\rho = \rho\sigma$, contrary to the chosen transpositions. For example, they send the symbol 0 to different symbols in the two orders. $\square$

Theorem 4.2 (Qualitative binary obstruction). *For $3 \leq k \leq n$ and arbitrary a, b, τ , equality (1.1) is impossible.*

Proof. If τ is not surjective, every coloring in its orbit omits at least one of the original missing labels. Thus assume τ is surjective.

Both a and b are permutations. Every orbit coloring has rank k , so no constant coloring is reachable.

Both a and b are singular. Choose one collision pair for each and form a graph H consisting of the resulting one or two edges. This graph is 3-colorable, so [theorem 3.2](#) gives a proper surjective k -coloring. Every nonempty product ends on the right in a or b and collapses the chosen pair for that factor. Hence the only possibly reachable proper coloring is the initial coloring τ . There are at least two distinct proper colorings, obtained by swapping two used output colors, so one is missing.

Exactly one generator is a permutation. Write it as p and the singular one as s . Choose a proper surjective coloring c of $G(p; x, y)$. All three colorings in [theorem 4.1](#) are proper. If every coloring were reachable, [theorem 3.3](#) would put all three in the single cyclic orbit of τ under p . This contradicts [theorem 4.1](#). $\square$

The argument does not estimate the size of the transformation monoid. In particular, merely knowing that two transformations cannot generate all of $T(Q)$ would not have sufficed: a proper submonoid could in principle still meet every fiber of the map $t \mapsto \tau \circ t$. The graph argument excludes this possibility directly in the output-coloring space.

5 Second proof of nonattainment: counting the missing colorings

Second of two independent proofs. This section proves the quantitative inequality (1.2). What it buys over [section 4](#): a numerical additive deficit $k! - g(k)$ valid uniformly in n , which the qualitative route does not produce, and the lemma that generalizes to arbitrary permutation groups in [theorem 8.1](#). This is the only place where the shared core theorem is proved; every later bound refines it rather than reproving it.

5.1 A cyclic orbit sees only a cyclic subgroup of relabelings

Lemma 5.1 (Relabeling intersection bound). *Let $c : Q \rightarrow \Delta$ be surjective. If O is any orbit under precomposition by a permutation p , then*

$$|O \cap \mathcal{R}(c)| \leq g(k).$$

Proof. If the intersection is empty, there is nothing to prove. Otherwise choose $d \in O \cap \mathcal{R}(c)$. Then $O = \{d \circ p^j : j \in \mathbb{Z}\}$ and $\mathcal{R}(d) = \mathcal{R}(c)$. Define

$$K = \{j \in \mathbb{Z} : d \circ p^j = \sigma \circ d \text{ for some } \sigma \in \text{Sym}(\Delta)\}.$$

For each $j \in K$, surjectivity makes σ unique; denote it by $\phi(j)$. We claim that K is a subgroup of $\mathbb{Z}$ and that $\phi : K \rightarrow \text{Sym}(\Delta)$ is a homomorphism.

The exponent 0 is in K . If $r, t \in K$, then

$$d \circ p^{r+t} = \phi(r) \circ d \circ p^t = \phi(r) \circ \phi(t) \circ d.$$

Hence $r + t \in K$ and $\phi(r + t) = \phi(r)\phi(t)$. Also, $d \circ p^r = \phi(r) \circ d$ implies $d \circ p^{-r} = \phi(r)^{-1} \circ d$, so $-r \in K$.

Every subgroup of $\mathbb{Z}$ is cyclic. Its image $\phi(K)$ is therefore a cyclic subgroup of $\text{Sym}(\Delta)$ and has at most $g(k)$ elements. The colorings of $O \cap \mathcal{R}(d)$ correspond one-to-one to these image elements. Thus the desired intersection has size at most $g(k)$. $\square$

The statement is about distinct output relabelings, not about the number of exponents realizing them. Many powers of p may induce the same coloring. This is why the argument uses the image of a subgroup rather than the order of p itself.

It is convenient to name the cyclic orbit of an arbitrary output map:

$$\text{Orb}_\tau = \{\tau \circ p^j : j \in \mathbb{Z}\}. \quad (5.1)$$

With this notation theorem 5.1 reads $|\text{Orb}_\tau \cap \mathcal{R}(c)| \leq g(k)$ for every surjective c .

Remark 5.2 (A second packaging of the same bound: the setwise stabilizer). The same inequality can be obtained inside the finite cyclic group $A = \langle p \rangle$ rather than inside $\mathbb{Z}$. Take $\tau = c$ first. The nonempty fibers of c form an unordered partition $\mathcal{B}_c = \{c^{-1}(d) : d \in \Delta\}$. Let $H \leq A$ be its *setwise* stabilizer: those elements of A that permute the blocks, without necessarily preserving each block individually. For $h \in H$ there is a unique $\rho(h) \in \text{Sym}(\Delta)$ with

$$c \circ h = \rho(h) \circ c, \quad (5.2)$$

uniqueness coming from surjectivity of c ; conversely any $h \in A$ satisfying (5.2) for some color permutation lies in H . The same composition computation as above makes $\rho : H \rightarrow \text{Sym}(\Delta)$ a homomorphism, and $|\text{Orb}_c \cap \mathcal{R}(c)| = |\rho(H)|$. A subgroup of a cyclic group is cyclic, so $\rho(H)$ is a cyclic subgroup of $\text{Sym}(\Delta)$ and has at most $g(k)$ elements. For general τ , if the intersection is nonempty choose $c' = \pi_0 \circ c$ in it; then $\text{Orb}_\tau = \text{Orb}_{c'}$, and relabeling by π_0 bijects Orb_c with $\text{Orb}_{c'}$ while mapping $\mathcal{R}(c)$ onto itself, so the two intersections have equal size.

What this packaging buys over theorem 5.1: it is the form that generalizes verbatim to an arbitrary permutation group in theorem 8.1, and it is the form whose characteristic failure mode (the partition is stabilized setwise, not block by block) is flagged in section D. The $\mathbb{Z}$ -version is kept as the working lemma because choosing the base point d inside the intersection absorbs the reduction from a general τ that the finite-group version has to perform separately. The two are the same mathematics in different dress, and the different dress does different work.

Lemma 5.3. For $k \geq 3$, $g(k) < k!$.

Proof. A permutation of order $k!$ would generate a cyclic subgroup of the same size as $\text{Sym}(\Delta)$, hence all of it. But $\text{Sym}(\Delta)$ is not cyclic for $k \geq 3$: the transpositions of the first two and of the last two of three distinct symbols do not commute. $\square$

5.2 Proof of the quantitative theorem

Proof of theorem 1.1. We bound the number of missing colorings in each case.

If τ is not surjective, choose any surjective coloring $c : \mathcal{Q} \rightarrow \Delta$. All $k!$ members of $\mathcal{R}(c)$ are absent from $\tau\langle a, b \rangle$.

Suppose τ is surjective and both generators are permutations. Choose a coloring c using exactly $k - 1$ labels. None of its relabelings is reachable, since rank is preserved. Its stabilizer in $\text{Sym}(\Delta)$ is trivial: a permutation fixing the $k - 1$ used symbols must fix the one remaining symbol as well. Thus again at least $k!$ colorings are missing.

If both generators are singular, use the graph H from the qualitative proof and a proper surjective coloring c . All $k!$ colorings in $\mathcal{R}(c)$ are proper, but at most the initial coloring τ is reachable. Hence at least $k! - 1$ are missing.

Finally, in the mixed case choose a proper surjective coloring c of the collision-orbit graph. Its $k!$ relabelings are proper. By theorems 3.3 and 5.1, at most $g(k)$ of them belong to the reachable set. At least $k! - g(k)$ are therefore missing.

Every case gives at least $k! - g(k)$ missing colorings, since $g(k) \geq 1$. This proves the first inequality in (1.2). The strict inequality is theorem 5.3. Applying the reversal construction proves the automata assertion. $\square$

k	$k!$	$g(k)$	Guaranteed missing colorings $k! - g(k)$
3	6	3	3
4	24	4	20
5	120	6	114
6	720	6	714
7	5 040	12	5 028
8	40 320	15	40 305
9	362 880	20	362 860
10	3 628 800	30	3 628 770

Table 1: An additive deficit valid for every $n \geq k$, computed exactly from (1.3). It is not asserted to be the optimal deficit except at $(n, k) = (3, 3)$. The accompanying file `results/landau_gaps.json` continues the table to $k = 12$, where $g(11) = 30$ with deficit 39 916 770 and $g(12) = 60$ with deficit 479 001 540.

The universal gap is deliberately simple rather than usually sharp. It ignores all but one relabeling orbit of proper colorings. Counting the entire proper-coloring set will give substantially stronger bounds in section 9.

Remark 5.4 (Why three output labels matter). For two colors the counting deficit vanishes, since $g(2) = 2! = 2$. Independently, some collision-orbit graphs are odd cycles and have no proper 2-coloring. Both halves of the obstruction genuinely change at three colors. This is not a hidden argument that would incorrectly prohibit binary-alphabet reversal witnesses for ordinary languages; section 7.1 exhibits a $k = 2$ machine reaching all 2^n colorings.

Finally, a caution about what has been bounded. The proof does not classify two-generated monoids, estimate their orders, or enumerate their elements. Its bound concerns the action on output maps, which may be substantially smaller than the monoid itself. The distinction is especially relevant when $k < n$, where different transformations can have exactly the same composition with τ .

6 Constructing a missing reversal state

The qualitative proof yields a useful algorithm. Its input is two explicit transition arrays and an output array. Its output is a particular coloring that cannot be a reversal state, together with a short explanation that can be checked without exploring all k^n colorings.

6.1 The case split

If an output label does not occur in τ , return the constant coloring with that label. If both letters are permutations, every reachable coloring has the same number of occurrences of each label as τ , so any constant coloring with a different label-count vector is unreachable; deterministically, return the constant zero coloring, unless τ is already constantly zero, in which case return the constant one coloring.

If both letters are singular, find one collision pair for each. Properly color their one- or two-edge graph, split classes until k labels are used, and choose this coloring unless it equals τ . In that exceptional case, swap two output labels. The resulting proper coloring is not the initial state, and no nonempty word can reach it.

In the mixed case, form $G(p; x, y)$ by iterating the unordered pair under p . At most $\binom{n}{2}$ different pairs can occur, so the graph is built in polynomial time even when p has very large order. Construct a proper surjective coloring c : try a bipartite coloring first, and if the graph is not bipartite, the proof of theorem 3.1

shows that every nonisolated component is a cycle and every vertex has degree at most two, so greedy coloring uses at most three colors. Then consider

$$c_0 = c, \quad c_1 = (0\ 1) \circ c, \quad c_2 = (1\ 2) \circ c.$$

Test these colorings for membership in the cyclic orbit of τ . At least one test must fail by theorem 4.1. Since each candidate is proper, a failed cyclic-orbit test certifies that it is unreachable by any word.

Remark 6.1 (The counting route gives a six-candidate variant). If one starts instead from theorem 5.1 rather than from theorem 4.1, the natural search tests all six relabelings of a proper coloring that uses exactly the three labels 0, 1, 2: at most $g(3) = 3$ of them can lie in Orb_τ , so at most six tests are needed. More explicitly, if $\text{im}(\tau)$ is not exactly $\{0, 1, 2\}$ then the cyclic orbit contains none of the six, and otherwise the lemma applies directly. Both variants are implemented and both are correct; the three-candidate version is the cheaper one, and it is available only because section 4 was proved separately. This is one concrete payoff of keeping the two proofs apart.

The only potentially nontrivial computational issue is deciding

$$c = \tau \circ p^j \quad \text{for some integer } j \tag{6.1}$$

without iterating through all powers of p . The next construction resolves it with modular arithmetic.

6.2 A rotation constraint for each permutation cycle

Decompose p into cycles. For one cycle

$$C = (q_0, q_1, \dots, q_{m-1}), \quad p(q_i) = q_{i+1 \bmod m},$$

form the two cyclic words

$$U_i = \tau(q_i), \quad V_i = c(q_i).$$

Equation (6.1) requires

$$V_i = U_{i+j \bmod m} \quad (0 \leq i < m).$$

If V is not a cyclic rotation of U , the candidate is immediately rejected. Otherwise find one working shift r , and let d be the least positive cyclic period of U :

$$U_{i+d \bmod m} = U_i \quad \text{for all } i.$$

Here $d \mid m$. Every working shift satisfies exactly the congruence

$$j \equiv r \pmod{d}. \tag{6.2}$$

Lemma 6.2. *The shifts fixing a cyclic word of length m form a subgroup of $\mathbb{Z}/m\mathbb{Z}$. If a target word is one rotation of the source, the shifts taking the source to the target form a coset of this subgroup. Equivalently, they are precisely one residue class modulo the least cyclic period.*

Proof. The identity shift fixes the word; the composition and inverse of fixing shifts also fix it. Thus the fixing shifts form a subgroup. If r takes the source to the target, another shift j does so exactly when $j - r$ fixes the source. Subgroups of $\mathbb{Z}/m\mathbb{Z}$ consist of the multiples of a divisor of m , and the least positive period is that divisor. $\square$

For example, the word 010101 has least cyclic period 2, not 6. A target 101010 contributes the constraint $j \equiv 1 \pmod{2}$. Using the whole cycle length as the modulus would incorrectly discard valid shifts.

6.3 Combining the constraints

Maintain a congruence $j \equiv R \pmod{M}$, initially $R = 0$, $M = 1$. To add $j \equiv r \pmod{d}$, put $g = \gcd(M, d)$. The two congruences are compatible if and only if

$$r - R \equiv 0 \pmod{g}.$$

When they are compatible, write $j = R + Mt$. Dividing by g gives

$$\frac{M}{g}t \equiv \frac{r - R}{g} \pmod{d/g}.$$

The coefficient M/g is invertible modulo d/g , so this determines a unique class of t and hence one class of j modulo $\text{lcm}(M, d)$. Modulus 1 is handled as a vacuous congruence.

A rejection certificate is either a cycle whose target is not a rotation of the source, or a pair of incompatible accumulated congruences. A successful test returns an exponent representing the solution class. At most three such tests are needed to produce an unreachable proper coloring.

6.4 A second, deliberately different membership criterion

The incremental merge just described is not the only way to decide the system. Suppose the cycles contribute congruences $j \equiv r_i \pmod{p_i}$ for $i = 1, \dots, v$. The generalized Chinese remainder theorem says that the system is solvable exactly when every pair is compatible:

$$r_i \equiv r_{i'} \pmod{\gcd(p_i, p_{i'})} \quad \text{for all } i, i'. \quad (6.3)$$

Sufficiency follows by prime powers. For each prime, select a congruence whose modulus carries the largest exponent of that prime. Pairwise compatibility makes its residue agree with every weaker requirement at that prime. The selected prime-power congruences have coprime moduli and combine by the ordinary Chinese remainder theorem, and their common solution satisfies the original system.

What this buys over the incremental merge: nothing mathematically, and that is the point. The two criteria are used by two programs that share no code, so a disagreement between them exposes an implementation mistake in one of them. The producer code/dfao.py merges congruences iteratively; the standalone verifier code/certificate_checker.py uses (6.3). This deliberate divergence of method is the substance of the trust separation described in section 6.7, and the two implementations are not unified.

Proposition 6.3 (Polynomial-time constructive obstruction). *For explicit arrays a, b, τ with $3 \leq k \leq n$, one can construct a coloring outside $\tau\langle a, b \rangle$ in time polynomial in the input size, without enumerating the coloring orbit or the transformation monoid. The construction also admits a polynomial-size certificate that can be checked in polynomial time.*

Proof. Finding collisions and decomposing a permutation into cycles are elementary finite-array operations. The graph has at most $\binom{n}{2}$ edges and can be built in polynomial time. It is either bipartite or has maximum degree at most two, so the initial proper coloring is obtained by a breadth-first bipartite coloring or a greedy 3-coloring. Splitting color classes takes at most k further steps.

The simple implementation tests all rotations of each cycle word, requiring $O(\sum m_i^2) \leq O(n^2)$ symbol comparisons for one cyclic-orbit test. The least periods can be found within the same bound. The moduli in the Chinese remainder calculation divide $\text{lcm}(1, 2, \dots, n)$ and therefore have at most $O(n \log n)$ bits, using the elementary bound by $n!$. Euclidean arithmetic and modular inverses on these integers take polynomial time. Only three candidate tests are needed.

For the certificate clause, a certificate consists of the input arrays and the target, together with either a differing label-count vector, or two collision pairs, or one collision pair with its orbital graph and evidence

of cyclic nonmembership. In the last case the evidence is a cycle whose target word is no rotation of the source word, or a pair of incompatible period congruences; a checker may instead recompute all congruences and reject membership directly. The graph has at most $\binom{n}{2}$ edges and every tuple has length at most n , so both the certificate size and the checking time are polynomial. $\square$

6.5 A concrete certificate

Consider the 8-state, 5-output machine whose arrays are

$$\begin{aligned} p &= (1, 2, 0, 4, 5, 6, 7, 3), \\ s &= (3, 2, 1, 3, 4, 5, 6, 0), \\ \tau &= (0, 1, 2, 3, 4, 4, 4, 4). \end{aligned}$$

Array notation means, for example, $p(0) = 1$. The collision $s(0) = s(3)$ generates $K_{3,5}$ between $\{0, 1, 2\}$ and $\{3, 4, 5, 6, 7\}$. The algorithm returns

$$c = (2, 3, 0, 4, 1, 1, 1, 1).$$

Its two block palettes, $\{0, 2, 3\}$ and $\{1, 4\}$, are disjoint, so it is proper. On the first cycle, however, $(2, 3, 0)$ is not a rotation of the source word $(0, 1, 2)$. Thus c is not a power-rotation of τ , and no word containing s can reach a proper coloring. This proves that c is missing. The full machine-readable certificate is in `data/example_certificate.json`. Its schema is

```
{n, k, a, b,  $\tau$ , unreachable{case, target, permutation, singular,
                                collision, edges, candidate_tests[...]}},
```

recording the 15 edges of $K_{3,5}$ and, for each of the three candidates, the outcome of its cyclic-orbit test with reason `nonrotation`.

6.6 Two certificate formats and two generators

The archive ships a second, differently shaped certificate family, produced by a different search and validated by a different program. Its schema is

```
{n, k, a, b,  $\tau$ , case, permutation_letter, pair, edges, target, cyclic_nonmembership{...}},
```

in `results/certificates.json`, where the nonmembership block records `member`, `reason`, `cycle` and `constraints`. Four such certificates are supplied, all in the one-permutation case: at $(n, k) = (3, 3)$ with target $[0, 2, 1]$; at $(4, 3)$ and $(4, 4)$ with target $[2, 1, 0, 1]$; and at $(5, 3)$ with target $[2, 0, 1, 1, 1]$. All four carry the reason `no_rotation`.

The two formats are kept side by side and neither schema is normalized onto the other. What each buys: the `results/certificates.json` family is produced by the six-relabeling search of theorem 6.1 and is checked by the import-free verifier described below, which is what makes it useful as a trust artifact, and its structural variant scales to $n \leq 50$ without any orbit enumeration; the `data/example_certificate.json` family is produced by the three-candidate search driven by theorem 4.1, records the candidate-by-candidate test log that the cheaper search generates, and describes a single large machine, the 8-state, 5-output $K_{3,5}$ example whose orbit is computed exactly in section 10. Different search strategies and different scales make agreement between them informative.

6.7 Implementation and trust separation

The file `code/dfao.py` constructs the certificates of the first family. The separate file `code/certificate_checker.py` imports no code from that module. It recomputes the collision conditions, the entire pair orbit, properness, and cyclic nonmembership, using the pairwise criterion (6.3) rather than the producer's iterative merge, and exits with a nonzero status if any supplied certificate is rejected.

This independence is limited, and is stated precisely. Both programs implement the same mathematical argument, so they are not an independent proof of the theorem. They are independent checks of finite certificates under that argument. The archive also includes randomized comparisons with explicit cyclic-orbit enumeration and direct comparisons with complete reverse-state searches.

7 Why three input letters suffice

The binary obstruction is an exact alphabet-size threshold, not an obstruction to maximal reversal in general. We include a construction to make this distinction explicit.

Theorem 7.1 (Ternary attainability). *For every $3 \leq k \leq n$, some minimal n -state DFAO with three input letters and k output values has reversal state complexity k^n .*

Proof. Let $Q = \{0, \dots, n-1\}$. Use three transformations:

$$p = (0\ 1 \cdots n-1), \quad t = (0\ 1),$$

and the map e that sends 1 to 0 and fixes every other point. Conjugating t by powers of p yields the adjacent transpositions, so p and t generate $\text{Sym}(Q)$.

Conjugates of e by permutations give every elementary collapse that sends one chosen point to another and fixes all remaining points. These collapses and the permutations generate every transformation of Q . To see this, let $f : Q \rightarrow Q$ be arbitrary. Choose one representative $r_y \in f^{-1}(y)$ for each $y \in \text{im}(f)$. Collapse every other point of $f^{-1}(y)$ onto r_y . The resulting transformation h satisfies $h(q) = r_{f(q)}$. The bijection $r_y \mapsto y$ between two equally sized subsets extends to a permutation u of Q . Then $f = u \circ h$.

Choose any surjective output map $\tau : Q \rightarrow \Delta$. Given any coloring $c : Q \rightarrow \Delta$, choose a representative $r_d \in \tau^{-1}(d)$ for every output symbol d , and define $f_c(q) = r_{c(q)}$. Then $\tau \circ f_c = c$. Since the transition monoid is all of $T(Q)$, every coloring is reachable in the reversal construction. The original machine is accessible because it contains the n -cycle p . Hence theorem 2.1 gives k^n distinguishable reversal states.

The original machine is also minimal. If $q \neq q'$, choose states u, v with different outputs. A transformation sending q to u and q' to v is realized by an input word, which distinguishes q from q' . $\square$

Combining this theorem with theorem 1.1 proves the second assertion of theorem 1.2: the least input-alphabet size that permits all k^n reversal states is exactly three for $3 \leq k \leq n$. A one-letter machine is covered by duplicating its transformation.

Remark 7.2 (The two-value case). One of the two source reports stated this attainability for every $n \geq k \geq 2$. Its construction uses only that p and t generate $\text{Sym}(Q)$ and that elementary mergers together with permutations generate all of $T(Q)$, neither of which needs $k \geq 3$, so the case $k = 2$ is covered on the same grounds. The deficit of theorem 1.1 is a separate matter and is claimed only for $k \geq 3$.

Ternary attainability was already established in the target paper [1, Theorem 1 and Corollary 2]; the verification above only fixes conventions, and the new obstruction supplies the missing binary exclusion.

7.1 Why two output values are different

The proof genuinely needs three output values. A collision-orbit graph can contain an odd cycle and then has no proper 2-coloring. Independently, $\text{Sym}(\{0, 1\})$ is cyclic, so the noncommuting-relabellings argument is unavailable even when a proper binary coloring exists.

A small explicit example uses

$$p = (1, 2, 0), \quad s = (0, 0, 2), \quad \tau = (0, 0, 1).$$

The orbit contains all eight binary colorings of three states. Powers of p provide the three colorings with one 1. Applying s to a suitable one provides a coloring with two 1s, whose rotations give the other two. The same collapse produces 000 from a one-1 coloring and 111 from a two-1 coloring. Thus the small example already shows that the conclusion cannot simply be extended to $k = 2$.

The cases $k = 1$ and $k > n$ are outside the interesting range. The former gives a constant function; in the latter, no output map can use all k labels, and missing output values already prevent a full coloring orbit.

8 An extension to abelian permutation groups

The cyclic nature of the single permutation letter can be weakened. The same mechanism applies to any number of commuting permutation letters, provided there is only one singular letter.

8.1 The group-theoretic form of the intersection lemma

Proposition 8.1 (Relabelings induced by a permutation group). *Let $H \leq \text{Sym}(Q)$ and let $c : Q \rightarrow \Delta$ be surjective. If O is an H -orbit of colorings, then the nonempty intersection $O \cap \mathcal{R}(c)$ has size $|J|$ for a subgroup $J \leq \text{Sym}(\Delta)$ that is a homomorphic image of a subgroup of H .*

Proof. Choose $d \in O \cap \mathcal{R}(c)$, so $O = dH$ and $\mathcal{R}(d) = \mathcal{R}(c)$. Define

$$K = \{h \in H : d \circ h = \sigma \circ d \text{ for some } \sigma \in \text{Sym}(\Delta)\}.$$

The output permutation is unique. The same composition and inverse calculation as in theorem 5.1 shows that K is a subgroup of H and that the assignment $h \mapsto \sigma$ is a homomorphism. Its image J parametrizes exactly the colorings in the intersection. $\square$

Thus a full output-relabeling orbit would force $\text{Sym}(\Delta)$ to be a quotient of a subgroup of H . This is a general obstruction, although the graph colorability condition must still be checked.

Remark 8.2 (Several singular generators). Nothing above needs a single singular letter. Let $H \leq \text{Sym}(Q)$ be generated by any set of permutation letters, let $b_1, \dots, b_r$ be singular letters, and choose one collision pair e_i for each b_i . Put

$$E = \bigcup_{i=1}^r \{h(e_i) : h \in H\}. \quad (8.1)$$

A composition containing a singular letter has a rightmost such letter and a suffix in H , so exactly as in theorem 3.3 its output coloring is improper for this graph, and the reachable proper colorings lie in τH . If that graph has a proper surjective k -coloring c , taking H to be the setwise stabilizer of the fiber partition of c as in theorem 5.2 shows that full reachability forces $\rho(H) = \text{Sym}(\Delta)$.

The colorability hypothesis cannot simply be dropped. With many collision pairs or a large permutation group, the graph of (8.1) can require more than k colors. Moreover, a nonabelian permutation group can genuinely have a symmetric-group quotient. These two escape routes are exactly why the extension does not contradict the three-letter full-transformation witness of theorem 7.1.

8.2 Orbit graphs for abelian groups

For a collision pair $s(x) = s(y)$, define

$$E(G_H) = \{\{h(x), h(y)\} : h \in H\}.$$

A word containing s can again be written $u \circ s \circ h$ with $h \in H$. It collapses an edge of G_H , so reachable proper colorings must lie in the single H -orbit τH .

Lemma 8.3. *If H is abelian, the graph G_H is 3-colorable.*

Proof. If x and y lie in different H -orbits, all edges join those two orbits, so the graph is bipartite. Otherwise write $y = h_0(x)$ for some $h_0 \in H$. For a vertex $z = h(x)$, commutativity gives

$$h(y) = h(h_0(x)) = h_0(h(x)) = h_0(z).$$

Thus the edges on the orbit of x are exactly $\{z, h_0(z)\}$. Every vertex has at most two neighbors. All other vertices are isolated, so three colors suffice. $\square$

Let $A(k)$ denote the largest order of an abelian subgroup of $\text{Sym}(k)$. There is a simple exact expression:

$$A(k) = \max_{\lambda \vdash k} \prod_i \lambda_i. \quad (8.2)$$

To prove the upper bound, decompose an abelian permutation group into its orbits of sizes λ_i . On one orbit, an element fixing one point fixes every point, because it commutes with the transitive action. The faithful image on that orbit is therefore regular, of order λ_i . The original group embeds into the product of these images, so its order is at most $\prod_i \lambda_i$. Conversely, disjoint cyclic groups of orders λ_i attain that product.

For $k \geq 2$, the usual elementary product maximization gives

$$A(k) = \begin{cases} 3^{k/3}, & k \equiv 0 \pmod{3}, \\ 4 \cdot 3^{(k-4)/3}, & k \equiv 1 \pmod{3}, k \geq 4, \\ 2 \cdot 3^{(k-2)/3}, & k \equiv 2 \pmod{3}. \end{cases}$$

Indeed, a part at least 5 is improved by replacing it with 3 and the remaining part; parts 1 are eliminated in a maximizing partition for $k \geq 2$; and three parts 2 are improved by two parts 3. A leftover 4 may be kept as 4 or split into 2 + 2.

Theorem 8.4 (Abelian-permutation obstruction). *Let $H \leq \text{Sym}(Q)$ be abelian, let $s : Q \rightarrow Q$ be singular, and let $3 \leq k \leq n$. Then, for every $\tau : Q \rightarrow \Delta$,*

$$|\tau\langle H, s \rangle| \leq k^n - k! + A(k) < k^n.$$

Proof. A nonsurjective τ omits all $k!$ relabelings of any surjective coloring. Otherwise, theorems 3.2 and 8.3 gives a proper surjective k -coloring of G_H . Its reachable relabelings must lie in τH . By theorem 8.1, those relabelings form a set whose size is the order of an abelian subgroup of $\text{Sym}(\Delta)$, and so is at most $A(k)$. The remaining $k! - A(k)$ relabelings are missing. Since $\text{Sym}(k)$ is nonabelian for $k \geq 3$, $A(k) < k!$. $\square$

This does not forbid three-letter maximal examples in general: the two permutation letters in theorem 7.1 generate a nonabelian group. It instead identifies a structural requirement on any such construction with one singular letter.

9 Sharper bounds from all proper colorings

The graph-coloring viewpoint has antecedents in Holzer and König’s work on transformation monoids [2, Sections 4–5]; the coloring count is not presented here as a newly invented technique. The factorial gap counts one free output-relabeling orbit. We now count all proper colorings of a collision-orbit graph. The resulting bound retains information about the cycle structure of the permutation and is often sharp.

9.1 The basic chromatic bound

Let $P_G(k)$ be the number of proper maps $Q \rightarrow \Delta$. In the mixed case, theorem 3.3 immediately gives

$$|\tau\langle p, s \rangle| \leq k^n - P_G(k) + \text{ord}(p). \quad (9.1)$$

More generally, $\text{ord}(p)$ may be replaced by any bound on the p -orbit length of a proper coloring. If τ is improper, no proper coloring is reachable at all; using the larger term remains a valid upper bound.

Replacing $\text{ord}(p)$ by the actual orbit size, and splitting on properness rather than absorbing it, gives a strictly sharper statement for a given machine.

Proposition 9.1 (Chromatic deficit, instance-specific form). *Let p be a permutation, let s be singular, fix a collision pair $\{x, y\}$ of s , and put $G = G(p; x, y)$ and $M = \langle p, s \rangle$. Then*

$$|\tau M| \leq k^n - P_G(k) + \begin{cases} |\text{Orb}_\tau|, & \tau \text{ is proper for } G, \\ 0, & \tau \text{ is not proper for } G. \end{cases} \quad (9.2)$$

Proof. The permutation p is a graph automorphism of G , since $E(G)$ was defined in (3.1) as a p -invariant set of pairs. Its pullback action therefore preserves properness, so the orbit Orb_τ of (5.1) is wholly proper or wholly improper according to whether τ is proper. By theorem 3.3, every reachable proper coloring lies in that orbit. There are $k^n - P_G(k)$ improper colorings in total, and at most the stated number of reachable proper ones. Adding the two counts gives (9.2). $\square$

Remark 9.2 (Two instance-specific chromatic bounds, and what each buys). Inequalities (9.1) and (9.2) are both kept, and neither substitutes for the other. Theorem 9.1 is sharper for any fixed machine, because $|\text{Orb}_\tau|$ divides $\text{ord}(p)$ and is usually much smaller, and because an improper τ contributes nothing at all; it is what yields the exact values 24, 67, 176 and 215 in section 9.5. Inequality (9.1) is weaker for a given instance but is stated in terms of $\text{ord}(p)$ alone, and that is precisely what allows it to be maximized over all collision graphs without reference to τ ; it is the form that abstracts into the residual-order function $H_r(t)$, the period improvement $T_{r,k}(a, b)$, and hence the universal bound $U(n, k)$ of theorem 9.4. The instance-specific version cannot play that role, since $|\text{Orb}_\tau|$ is not a function of the graph parameters.

Neither inequality asserts that all improper colorings are reachable. That additional saturation property must be proved or checked separately for any claimed equality witness, which is why a graph bound alone does not solve the exact state-complexity optimization problem.

There are only two structural forms for $G(p; x, y)$.

9.2 A pair in one permutation cycle

Suppose the pair lies on a cycle of length m , with displacement h . Put

$$d = \gcd(m, h), \quad L = m/d, \quad r = n - m.$$

The graph consists of d disjoint cycles of length L , together with r isolated vertices. When $L = 2$, each component is one edge rather than a cycle with two parallel edges; the coloring count is unchanged.

For $L \geq 2$, the appropriate cycle factor is

$$C_L(k) = (k-1)^L + (-1)^L(k-1). \quad (9.3)$$

One direct derivation uses the $k \times k$ matrix B with zeros on the diagonal and ones elsewhere. A proper coloring of a labeled cyclic sequence is a closed walk, so the count is $\text{tr}(B^L)$. The constant vectors have eigenvalue $k-1$, and the subspace of coordinate sum zero has eigenvalue -1 and dimension $k-1$. This gives (9.3). At $L = 2$ the formula is $k(k-1)$, the count for one edge.

Therefore

$$P_G(k) = C_L(k)^d k^r. \quad (9.4)$$

Every divisor $d \mid m$ with $d < m$ is possible: take displacement $h = d$.

9.3 A pair in different permutation cycles

Let the cycle lengths be a, b , and put

$$d = \gcd(a, b), \quad u = a/d, \quad v = b/d, \quad r = n - a - b.$$

Label the two cycles by $\mathbb{Z}/a\mathbb{Z}$ and $\mathbb{Z}/b\mathbb{Z}$, putting the chosen endpoints at their respective origins. An edge joins i to j exactly when a common power of p carries the origin pair to (i, j) . The generalized Chinese remainder criterion says this occurs precisely when $i \equiv j \pmod{d}$. Hence

$$G \cong dK_{u,v} \sqcup rK_1.$$

The proper-coloring count for $K_{u,v}$ is

$$B_k(u, v) = \sum_{i=1}^{\min(u,v)} \binom{k}{i} \left\{ \begin{matrix} u \\ i \end{matrix} \right\} (k-i)^v. \quad (9.5)$$

To derive it, let the left side use exactly i colors. Partition its u labeled vertices into i nonempty color classes in $\left\{ \begin{matrix} u \\ i \end{matrix} \right\}$ ways, assign distinct output labels to the classes in $\binom{k}{i} = k!/(k-i)!$ ways, and color each right-side vertex with one of the remaining $k-i$ labels. Thus

$$P_G(k) = B_k(u, v)^d k^r. \quad (9.6)$$

This derivation also proves the symmetry $B_k(u, v) = B_k(v, u)$, even though (9.5) does not display it symmetrically.

For three colors the expression simplifies considerably:

$$B_3(u, v) = 3(2^u + 2^v - 2), \quad u, v \geq 1. \quad (9.7)$$

Only $i = 1, 2$ contribute in (9.5), and $\left\{ \begin{matrix} u \\ 2 \end{matrix} \right\} = 2^{u-1} - 1$.

9.4 Choosing a collision pair

A singular map may have many collision pairs. Each one gives a valid bound of the form (9.2), so taking the minimum over all collision pairs of s is a legitimate improvement. Alternatively, the union of all the corresponding orbital graphs is still a valid obstruction graph, because a word containing s collapses one of its edges whichever pair is chosen. That union may, however, cease to be three-colorable, and its chromatic polynomial may be harder to compute. The uniform proof deliberately uses a single pair: that choice is what preserves the elementary three-colorability argument of theorem 3.1 for every permutation, and hence the guaranteed supply of surjective proper colorings.

9.5 Worked instances

Tuples list map values in the order $0, 1, \dots, n-1$. All orbit counts in this subsection are computed by exact integer-state breadth-first search.

Sharpness at three states and three outputs. Take

$$a = (1, 2, 0), \quad b = (0, 0, 2), \quad \tau = (0, 1, 2). \quad (9.8)$$

The collision pair $\{0, 1\}$ produces a triangle under the three-cycle a . There are six proper three-colorings, and the pure- a orbit of τ is $(0, 1, 2), (1, 2, 0), (2, 0, 1)$, of size three. Thus theorem 9.1 gives at most $27 - 6 + 3 = 24$ reachable states. The word table of section C exhibits 24 distinct colorings, so this automaton has exactly 24 reverse states and the three missing ones are

$$(0, 2, 1), \quad (1, 0, 2), \quad (2, 1, 0).$$

The original machine is accessible and minimal: the cycle reaches every state, and τ already separates states without a continuation. Hence equality holds in theorem 1.1 at $n = k = 3$.

Four-state examples. Use $a = (1, 2, 3, 0), b = (0, 0, 3, 2)$, again with the collision pair $\{0, 1\}$, whose orbital graph is the four-cycle C_4 . For three colors take $\tau = (0, 1, 0, 2)$: it is proper, its cyclic orbit has size four, and $C_4(3) = 2^4 + 2 = 18$ by (9.3), so the bound is $81 - 18 + 4 = 67$. For four colors take $\tau = (0, 1, 2, 3)$: now $C_4(4) = 3^4 + 3 = 84$ and the bound is $256 - 84 + 4 = 176$. Exact search attains both. Both machines are accessible and minimal.

A complete-bipartite example. Let

$$a = (1, 0, 3, 4, 2), \quad b = (2, 1, 2, 3, 0), \quad \tau = (0, 1, 2, 2, 2).$$

The permutation has cycles $(0\ 1)$ and $(2\ 3\ 4)$, while b identifies 0 and 2, so the orbital graph is $K_{2,3}$. By (9.5),

$$B_3(2, 3) = 3 \cdot 2^3 + 6 \cdot 1^3 = 30,$$

in agreement with (9.7). The coloring τ is proper with orbit size two under a , so theorem 9.1 gives $3^5 - 30 + 2 = 215$, which the included search attains.

Remark 9.3 (Two different $n = 5, k = 3$ examples). The value 215 just obtained is an *instance-specific* equality for this particular $K_{2,3}$ machine; it is not a claim that 215 is the global five-state, three-output maximum. Section A lists a different five-state machine, $p = (1, 2, 3, 4, 0), s = (0, 0, 3, 4, 2), \tau = (0, 1, 0, 1, 2)$, whose orbit has size $218 = U(5, 3)$ and which realizes the small-case maximum of section 10. The two examples illustrate different claims, the first that the cross-cycle formula is exactly attained and the second that the structural bound is met, and $215 < 218$ is a consequence of the first machine not being extremal. They are not in conflict, and neither replaces the other.

n	k	k^n	Uniform bound	Graph bound	Actual witness orbit
3	3	27	24	24	24
4	3	81	78	67	67
4	4	256	236	176	176
5	3	243	240	215	215

Table 2: The uniform estimate of theorem 1.1 is designed to prove nonattainment; the instance-specific graph estimate of theorem 9.1 can be much stronger for a particular pair of generators. The last row is the $K_{2,3}$ machine of theorem 9.3, not the extremal five-state one.

The values 24, 67 and 176 are not new numerical records: they occur among the previously verified maxima in [1, Table 3]. Here they serve as independent regression tests and as examples where the collision graph explains the missing states. For every example the archive records the input maps, the exact orbit size, a minimality check, and an explicit unreachable coloring; the certificate checker verifies nonreachability structurally, without relying on the breadth-first search that produced the orbit counts.

9.6 Accounting for the remaining permutation cycles

Let O_r be the set of orders of permutations on r points, with $O_0 = \{1\}$. Define

$$H_r(t) = \max_{o \in O_r} \text{lcm}(t, o). \quad (9.9)$$

For a same-cycle pair the distinguished cycle contributes order m , so

$$\text{ord}(p) \leq H_r(m).$$

For a different-cycle pair, the two distinguished cycles contribute $\text{lcm}(a, b)$, giving

$$\text{ord}(p) \leq H_r(\text{lcm}(a, b)).$$

The exact set of possible residual orders matters. In general one must not replace $H_r(t)$ by $\text{lcm}(t, g(r))$: the residual order maximizing an ordinary integer magnitude need not maximize its least common multiple with t .

For example, $g(5) = 6$. If $t = 6$, then $\text{lcm}(t, g(5)) = 6$, whereas a residual 5-cycle gives $\text{lcm}(6, 5) = 30$. The definition (9.9) handles such interactions correctly.

9.7 The three-color period improvement

When $k = 3$ and $\gcd(a, b) = 1$, the nonisolated part is the connected graph $K_{a,b}$. In a proper coloring the palettes on the two sides are disjoint. Since only three labels are available and both palettes are nonempty, at least one side is monochromatic. Its rotation contributes period 1. Thus the proper-coloring period on the two distinguished cycles divides a or divides b , rather than necessarily requiring $\text{lcm}(a, b)$. Including the isolated vertices gives the improved bound

$$T_{r,k}(a, b) = \begin{cases} \max\{H_r(a), H_r(b)\}, & k = 3, \gcd(a, b) = 1, \\ H_r(\text{lcm}(a, b)), & \text{otherwise.} \end{cases} \quad (9.10)$$

The connectedness condition is important. We do not apply the improvement component by component when $\gcd(a, b) > 1$, because the permutation may rotate the components as well as the colors within them.

9.8 A universal finite formula

The two unmixed generator cases also admit easy bounds. If both letters are permutations, the orbit lies in one fixed color histogram. The largest possible histogram class is the balanced multinomial coefficient

$$M(n, k) = \frac{n!}{(q!)^{k-z}((q+1)!)^z}, \quad n = qk + z, \quad 0 \leq z < k.$$

Balancing is optimal because replacing two counts $a \geq b + 2$ by $a - 1, b + 1$ decreases the product of their factorials.

If both letters are singular, their two chosen collision edges have at least $(k-1)^2 k^{n-2}$ proper colorings. For two distinct edges the count is exactly this whether they share a vertex or are disjoint; for a repeated edge the count is larger. At most the initial coloring is reachable among them. Thus the missing-coloring count is at least

$$(k-1)^2 k^{n-2} - 1.$$

Define $D(n, k)$ to be the minimum of the following finite collection:

$$k^n - (k-1)^n, \tag{9.11}$$

$$k^n - M(n, k), \tag{9.12}$$

$$(k-1)^2 k^{n-2} - 1, \tag{9.13}$$

$$C_{m/d}(k)^d k^{n-m} - H_{n-m}(m), \tag{9.14}$$

$$B_k(a/d, b/d)^d k^{n-a-b} - T_{n-a-b, k}(a, b). \tag{9.15}$$

In (9.14), the parameters range over $2 \leq m \leq n$ and positive divisors $d \mid m$ with $d < m$. In (9.15), they range over $1 \leq a \leq b$, $a + b \leq n$, with $d = \gcd(a, b)$. The first entry handles nonsurjective output maps. It is unnecessary when surjectivity is part of the definition of the extremal problem, but makes the bound valid for arbitrary supplied output arrays.

Theorem 9.4 (Structural upper bound). *For every $3 \leq k \leq n$ and arbitrary a, b, τ ,*

$$|\tau\langle a, b \rangle| \leq U(n, k) := k^n - D(n, k). \tag{9.16}$$

It is always valid to combine this with theorem 1.1 by taking the smaller of $U(n, k)$ and $k^n - k! + g(k)$.

Proof. Every machine belongs to one of the nonsurjective, two-permutation, two-singular, or mixed cases. The first three give (9.11)–(9.13). In the mixed case choose any collision pair. If its endpoints are on the same permutation cycle, use (9.4) and the order bound $H_{n-m}(m)$. Otherwise use (9.6) and (9.10). These are exactly the entries (9.14) and (9.15). The actual machine therefore misses at least one entry of this collection, and hence at least its minimum. $\square$

The bound is obtained without enumerating transformations. There are only quadratically many choices of the displayed graph parameters up to divisor bookkeeping. Computing the residual order sets involves integer partitions; we do not claim that this part is a polynomial-time algorithm in n . It is nevertheless a much smaller exact computation than enumerating all n^{2n} ordered pairs of transformations, let alone their output orbits.

10 Finite optimality results and explicit witnesses

10.1 Comparison with the published lower bound

For $3 \leq k < n$, define

$$L(n, k) = \max_{\substack{1 < \ell < m, \ell+m=n \\ \gcd(\ell, m)=1}} [k^n - B_k(\ell, m) + G_k(\ell, m)], \tag{10.1}$$

where

$$G_k(\ell, m) = \begin{cases} m, & k = 3, \\ \ell m, & k \geq 4. \end{cases}$$

Davies’s Corollary 3 provides accessible binary DFAO witnesses attaining this lower bound [1]. We use that result as a literature dependency for the finite optimality comparison, not for the principal obstruction theorem.

Let $R_2(n, k)$ denote the largest reversal complexity of an accessible n -state binary DFAO using k output values. Then

$$L(n, k) \leq R_2(n, k) \leq U(n, k).$$

For the displayed range $n \geq 7$, admissible coprime splits always exist. For example, take the neighboring halves for odd n ; for $n \equiv 0 \pmod{4}$ take $n/2 - 1, n/2 + 1$; and for $n \equiv 2 \pmod{4}$, $n \geq 10$, take $n/2 - 2, n/2 + 2$.

Proposition 10.1 (Computer-assisted finite equality). *Exact evaluation of (10.1) and (9.16) gives*

$$L(n, k) = U(n, k) \quad (7 \leq n \leq 30, 3 \leq k < n).$$

Consequently these values equal $R_2(n, k)$ throughout that finite range. There are $\sum_{n=7}^{30} (n - 3) = 372$ parameter pairs.

The complete exact-integer table is supplied as `data/finite_range_bounds.csv`. Each row gives the upper and lower values, a minimizing graph type, a maximizing coprime split, and the number of upper-bound candidates examined. The program compares the integers directly; there is no floating-point tolerance or probabilistic component. An independent checker reproduces all 372 pairs using an unrestricted composition recurrence for permutation orders, inclusion–exclusion for surjections, and nonzero cycle displacements rather than the primary implementation’s divisor enumeration.

The witnesses also have minimal original state count n . Indeed, the complete bipartite graph for an admissible split contains two disjoint edges, so

$$B_k(\ell, m) \leq (k - 1)^2 k^{n-2}.$$

Thus the witness reversal size is strictly larger than

$$k^n - (k - 1)^2 k^{n-2} = (2k - 1)k^{n-2} > k^{n-1}.$$

If its function had an original DFAO with fewer than n states, the general reversal construction would give at most k^{n-1} reversal states, a contradiction. Therefore the finite maxima apply equally when the original function is required to have state complexity exactly n .

n	k	Exact maximum in the checked range	k^n	Split (ℓ, m)
7	3	2 125	2 187	(3, 4)
7	4	15 472	16 384	(3, 4)
7	5	71 037	78 125	(3, 4)
8	3	6 452	6 561	(3, 5)
8	4	63 403	65 536	(3, 5)
8	5	369 020	390 625	(3, 5)
8	6	1 539 561	1 679 616	(3, 5)
9	3	19 550	19 683	(4, 5)
9	4	258 360	262 144	(4, 5)
10	3	58 654	59 049	(3, 7)

10.2 Small cases and the limitation at the diagonal

The structural upper bound also meets explicit breadth-first-search witnesses in nine of the ten pairs $3 \leq k \leq n \leq 6$.

n	k	Witness orbit size	Structural upper bound	Outcome
3	3	24	24	Equality
4	3	67	67	Equality
4	4	176	176	Equality
5	3	218	218	Equality
5	4	826	826	Equality
5	5	2 110	2 271	Bound not sharp
6	3	699	699	Equality
6	4	3 526	3 526	Equality
6	5	12 031	12 031	Equality
6	6	32 262	32 262	Equality

These are finite verifications: the upper bounds follow from a proved formula, and each listed lower witness is checked by exhaustive exploration of its coloring orbit. This is not exhaustive enumeration of all automata. At $(n, k) = (5, 5)$ the collision graph loses information and leaves a gap. The diagonal case $k = n$ is closely tied to the size of a two-generated transformation monoid, because a surjective output map is then bijective. For background on that monoid problem, see Holzer and König [2]. The graph bound alone does not recover all restrictions on such a monoid.

For example, at $n = 6$ the minimizing same-cycle graph consists of two triangles. Therefore its proper-coloring count is $[k(k-1)(k-2)]^2$, and the full-cycle order is 6. For $k = 3, 4, 5, 6$ the bound is

$$k^6 - [k(k-1)(k-2)]^2 + 6,$$

which gives exactly 699, 3526, 12031, and 32262.

10.3 An arithmetic correction to the source table

The $n = 8, k = 5$ entry in Table 2 of arXiv version 2 is printed as 368020 [1, p. 13]. Its own formula, evaluated at $(\ell, m) = (3, 5)$, gives instead

$$\begin{aligned}
 B_5(3, 5) &= 5 \cdot 4^5 + (5 \cdot 4) \cdot 3 \cdot 3^5 + (5 \cdot 4 \cdot 3) \cdot 2^5 \\
 &= 5120 + 14580 + 1920 = 21620, \\
 5^8 - B_5(3, 5) + 15 &= 390625 - 21620 + 15 = 369020.
 \end{aligned}$$

This is an arithmetic discrepancy in the displayed table, not a counterexample to the formula or to the lower-bound conjecture. The printed PDF page was inspected to distinguish it from an HTML extraction error.

The explicit 8-state witness in section 6 has exactly 369020 reachable colorings. Two independent breadth-first-search implementations, one storing Python tuples and one encoding colorings as base-5 integers in C++, return that same count. As additional comparison checks, the sums and bitwise exclusive-or values of all reached integer encodings are respectively

$$72076669650 \quad \text{and} \quad 128342.$$

These checksums are auxiliary consistency checks, not substitutes for the mathematical bound or the reachability computations.

11 Verification record and reproducibility

All computations described here were executed for this draft. The programs use exact integers, finite sets, and deterministic searches. The only randomness is the fixed-seed selection of test instances; no random sampling is used to establish the exhaustive maxima below.

11.1 Exhaustive small-state enumeration

For each $2 \leq n \leq 4$ and $2 \leq k \leq n$, a C++17 program enumerates all n^n transformations, every unordered pair of transformations with repetition, and one representative of every surjective output map up to renaming colors. The representatives are restricted-growth strings with exactly k blocks, of which there are $\begin{Bmatrix} n \\ k \end{Bmatrix}$. The number of searched instances is therefore exactly

$$\binom{n^n + 1}{2} \begin{Bmatrix} n \\ k \end{Bmatrix}. \quad (11.1)$$

Only two reductions are used: interchanging the names of the two input letters, which does not change the generated monoid, and relabeling all output colors, which bijects reverse orbits because it commutes with every pullback. No restriction to one permutation generator, no state-conjugacy representative, no accessibility or minimality requirement, and no conjecturally maximal monoid is imposed. Every orbit is explored to closure by breadth-first search, with colorings encoded as base- k integers and pullbacks precomputed.

n	k	Output partitions	Searched instances	Maximum orbit
2	2	1	10	4
3	2	3	1 134	8
3	3	1	378	24
4	2	7	230 272	16
4	3	6	197 376	67
4	4	1	32 896	176

Table 3: Complete searches with only input-letter interchange and output-color relabeling removed, 462 066 reduced instances in total. The $k = 2$ rows are boundary checks lying outside the hypotheses of theorem 1.1.

The Python implementation independently recalculates the orbit of each maximizing witness emitted by the C++ program, and verifies that some initial state makes the original automaton accessible and minimal; those witnesses therefore realize the listed orbit values as genuine reversal state complexities. The unrestricted maxima themselves rest on the C++ enumeration, which Python does not repeat for every four-state candidate. The per-row counts of maximizers in the reduced search, together with one explicit maximizing triple (a, b, τ) per row, are recorded in `results/exhaustive.json`.

11.2 What was actually checked

The archive includes executable source and the exact output records from the following checks.

Check	Instances
All ordered pairs of transformations and all output maps for $n = k = 3$	19 683
Cyclic membership: CRT algorithm versus literal orbit search, $n = 3, 4$ with $k = 3$	161 838
Random and mixed-case orbit searches, checking constructed missing states and both upper bounds	2 550
Collision graphs: proper coloring, closed count, direct enumeration, and inclusion–exclusion	1 362
Permutation orders from cycle partitions versus all permutations, $0 \leq n \leq 8$	46 234
Upper versus lower bound for $7 \leq n \leq 30$, $3 \leq k < n$	372
Explicit automaton witnesses checked by independent C++ orbit search	20

The exhaustive $n = k = 3$ test uses 3^3 choices for each of a , b , and τ , giving $27^3 = 19683$ triples. Its largest coloring orbit has size 24, agreeing with [theorem 1.1](#) exactly. Nonsurjective output maps are included, so that branch of the constructive algorithm is also exercised.

For the graph tests, the code enumerates permutations and collision pairs for $3 \leq n \leq 5$, constructs the actual edge set, and compares three ways of counting proper 3-colorings: direct enumeration, inclusion–exclusion over edges, and the formulas in [section 9](#). The two general counting procedures do not use the graph classification.

The random tests use the recorded seed 20260920. They are regression tests, not evidence for an unrestricted quantifier. The unrestricted statements are supported by the proofs; the tests are intended to expose orientation, period, and implementation mistakes.

11.3 A second audit suite at a different reduction level

A separately written suite repeats and extends these checks with different conventions, and its counts are recorded separately rather than merged into the table above.

Audit	Cases
All <i>ordered</i> three-state map pairs, all six output bijections	4 374
Two cyclic-membership implementations versus explicit rotation	1 000
Cyclic orbit/relabeling intersection versus $g(k)$	252
Orbital chromatic formulas versus all colorings, through $n = 5$	2 724
Random exact reverse-orbit and missing-certificate checks, $n \leq 8$	330
Larger structural certificate checks, $3 \leq n \leq 50$	1 440
C++ maximizing witnesses recomputed and checked in Python	6

Table 4: All listed checks passed. The larger certificate tests do not enumerate their reverse orbits.

Remark 11.1 (Two exhaustive $n = k = 3$ audits, at different reduction levels). The first row of [table 4](#) covers all $27^2 \cdot 6 = 4374$ ordered transformation pairs with all six output bijections, checking each against the bound 24 and verifying that each constructed certificate names a state genuinely missing from the complete orbit; the corresponding row of [table 3](#) uses the symmetry-reduced count 378. The first row of the table in the previous subsection instead uses the fully unreduced $27^3 = 19683$ triples, including nonsurjective output maps, which exercises the constructive algorithm’s nonsurjective branch as well. All three counts are kept, because the reduction assumptions are themselves part of what is being checked: a mistake in the letter-interchange or color-relabeling reduction would show up as a disagreement between them, and they agree on the maximum 24.

For the chromatic-formula audit, all permutations and all unordered pairs are tested for $n = 3, 4, 5$, and every coloring is counted for each of $k = 3, 4$. Testing $k = 4$ on three vertices is legitimate for those chromatic identities, although it lies outside the $k \leq n$ convention of [theorem 1.1](#). The random reverse-orbit checks use $k = 3, 4$ where applicable and balance the zero-, one-, and two-permutation cases. The larger certificate tests allow k up to eight and include nonsurjective output maps.

There are also ten focused unit tests, covering invalid inputs, small Landau values, the sharp three-state example, cyclic-shift obstructions, consistent and inconsistent Chinese remainder data, the cross-cycle and odd-cycle coloring formulas, certificate validation and certificate tampering, and the finite-search safety limit. One design rule is tested explicitly: a breadth-first search that reaches its configured state limit raises an error rather than returning an incomplete orbit as if it were exact.

11.4 What the experiments establish

The searches give exact finite cardinalities for their explicitly stated ranges and validate the implementations on additional instances. They provide useful evidence against composition-direction mistakes, forgotten constant colorings, improper treatment of repeated edges, or incompatible cycle rotations. They do not establish the universal theorem by induction, prove priority, or certify the correctness of arbitrary source code. The proof of [theorem 1.1](#) is independent of all of them.

11.5 Files in the archive

The main implementation is `code/reversal.py`. It contains ordinary function composition, coloring actions, literal orbit search, accessibility and minimality checks, the missing-coloring algorithm, cyclic membership via generalized CRT, chromatic counts, and the structural upper-bound evaluator. It uses only the Python standard library and supports Python 3.9 or later.

`code/run_checks.py` reproduces the recorded Python verification suite and the finite-range CSV. `code/orbit_count.cpp` is an independent C++17 implementation of coloring-orbit search. It consumes the plain-text fixtures in `code/bfs_cases.txt`. `code/check_bound_certificate.py` independently reevaluates the finite upper and lower bounds without importing the primary implementation; `code/check_large_example.py` repeats the 369020-state example using Python tuples. The C++ representation encodes c as $\sum_i c(i)k^i$, so it does not depend on Python tuple hashing or on any graph theorem.

The second implementation family is rooted at `code/dfao.py`, which contains its own validation, pullback, cycle and Landau routines, an exact reverse-orbit search that refuses to return a partial orbit when its `max_states` limit is reached, orbital-edge and properness tests, three-coloring and refinement, cyclic membership, missing-certificate generation, Stirling numbers, the bipartite and orbital chromatic counts, and an accessibility-and-distinguishability check. `code/certificate_checker.py` is the standalone verifier of [section 6.7](#), and imports nothing from `dfao.py`. `code/run_experiments.py` runs the audits of [table 4](#) and `code/test_dfao.py` the ten unit tests.

Remark 11.2 (Two independent C++ programs). The archive contains two C++17 programs, which do different jobs and are both kept. `code/exhaustive.cpp` performs the complete enumeration of all transformation pairs and output-partition representatives for $n \leq 4$, emitting the maxima and maximizers of [table 3](#); `code/orbit_count.cpp` performs single-instance breadth-first search over the fixtures of `code/bfs_cases.txt` with base- k integer encoding and sum and exclusive-or checksums, its largest case visiting 1 539 561 states. The first answers “what is the maximum over all small machines”; the second answers “exactly how large is the orbit of this one machine”, at sizes far beyond the first program’s reach. Their outputs are `results/exhaustive.json` and `data/cpp_orbit_counts.csv`.

Remark 11.3 (Two independent arithmetic paths, and two independent confirmations of 369020). The finite bounds of section 10 are computed twice by different arithmetic. `reversal.py`'s `structural_bound` uses divisor enumeration, the Stirling recurrence, and order sets built from nondecreasing cycle parts; `check_bound_certificate.py` re-derives all 372 upper and lower values using an unrestricted composition recurrence for permutation orders, inclusion–exclusion for surjections, and direct enumeration of all nonzero displacements. The independence of method is the evidence, so both programs and both result files, `data/finite_range_bounds.csv` and `data/independent_bounds_check.json`, are kept. In the same spirit the 369020-state example is confirmed twice, by the integer-encoded C++ search and by `check_large_example.py`'s Python tuple-set search, agreeing on the count and on the checksums 72076669650 and 128342; both runs and both records, `data/cpp_orbit_counts.csv` and `data/independent_python_bfs.json`, are what turn the correction of section 10.3 from an assertion into a verified one.

`data/small_witnesses.json` and `data/bfs_witnesses.json` contain explicit arrays. The data directory also contains the finite-range bounds, the constructive certificate, and verification summaries. The `results` directory contains the second suite's records: `certificates.json`, `exhaustive.json`, `examples.json`, `sharp_n3_orbit.json` (the word table of section C in machine-readable form), `landau_gaps.json`, `audit_summary.json`, which distinguishes exact orbit checks from larger certificate-only tests, the test and reproduction logs, `environment.txt`, and `pdf_checks.json`. The last is a rendering audit recording page count, absence of LaTeX warnings and of unresolved reference markers, absence of text outside the page box, enumerated external links, verified word certificates, rendering at 140 dpi, checked `cleveref` names, and full-page visual inspection; it was performed on the 19-page predecessor article from which the certificate and enumeration material is drawn, and has not been rerun against the merged text. Two provenance records are kept, and both are needed because they document different things. `source_audit.md` lists the seven numbered locations actually inspected in the source, the Holzer–König attribution points, and the dependency boundary of the proofs; `LITERATURE_SEARCH.md` together with `RESEARCH_STATUS.md` lists the inspected URLs, the conference DOI, the representative search queries, and the seven-item statement of what is not claimed. The first says what was read in the sources; the second says what was searched for and what is being asserted.

To reproduce the main checks from the archive root:

```
python code/run_checks.py --max-n 30
python code/check_bound_certificate.py
c++ -std=c++17 -O2 code/orbit_count.cpp -o orbit_count
./orbit_count < code/bfs_cases.txt
c++ -std=c++17 -O3 code/exhaustive.cpp -o exhaustive
./exhaustive 4 > results/exhaustive.json
python code/run_experiments.py
python code/certificate_checker.py results/certificates.json
python -m unittest discover -s code -p 'test_*.py' -v
```

The certificate checker prints `ACCEPTED 4 certificate(s)` and exits nonzero on any rejection. Do not invoke Python with `-O` or with `PYTHONOPTIMIZE` set: the audit scripts use assertions deliberately. On Windows, the compiled executable can be invoked as `.\orbit_count.exe`, with the input file supplied by the shell or an equivalent process-redirection command. The mathematics does not require a particular operating system.

To rebuild the article, run `latexmk -pdf -interaction=nonstopmode article.tex`, or `pdflatex article.tex` at least twice, or use the included Makefile. No bibliography processor is needed; the bibliography is contained in the TeX source. No checksum manifest accompanies the archive: re-running verification rewrites some generated files, including recorded elapsed times, so fixed hashes would be misleading.

11.6 Limits of the verification

No Lean, Coq, Isabelle, or other proof-assistant formalization is supplied. The code does not prove the general theorem by enumeration, and the finite comparison does not settle the exact-optimum conjecture beyond its checked range. The proof of the lower-bound existence theorem used in theorem 10.1 is a cited literature dependency. Conversely, none of those qualifications changes the quantifiers of the independently proved upper bounds: their proofs apply to every allowed n , k , and pair of transformations.

12 Further questions and conclusions

12.1 The remaining extremal inequality

The finite results suggest a sharply defined next problem. Prove, or refute,

$$U(n, k) = L(n, k) \quad (n \geq 7, 3 \leq k < n).$$

This is a sufficient route to the exact binary optimum. It is an extremal inequality over chromatic counts and residual permutation orders, rather than a search over all automata.

A proof would need to show that no same-cycle graph, disconnected bipartite graph, or graph with isolated vertices yields a smaller proper-coloring loss than a suitable connected $K_{\ell, m}$ with coprime block sizes. The residual-order term $H_r(t)$ and the special three-color period bound are part of the statement; discarding either can obscure the sharp value.

Failure of this numerical equality would not automatically disprove the original exact-optimum conjecture. The graph bound is only an upper bound; a minimizing graph need not be realized by a two-generated monoid that reaches every coloring not excluded by the graph. The gap at (5, 5) already illustrates this distinction.

12.2 Using more than one collision

The proof selects only one collision of the singular generator. A map of rank substantially below $n - 1$ has many collisions. Taking the union of all their permutation translates may exclude many more colorings. Its proper colorings would still be unreachable through words containing the singular letter. However, the union need not remain 3-colorable. This creates a tradeoff: additional collision information strengthens counting but can remove the convenient supply of surjective proper colorings.

For a fixed output size, a useful refinement would optimize over subsets of collision pairs whose orbit graph remains k -colorable. Such a refinement may improve the diagonal case $k = n$, or provide sharper instance-specific certificates even when it does not improve the universal maximum.

12.3 What is bounded, and what is not

The yes-or-no target is settled within this draft: the complete k^n coloring space cannot be reached by two generators when $k \geq 3$. That conclusion is conditional only on the correctness of the proof, not on a monoid-classification conjecture and not on the experimental data. Acceptance of the result as a new theorem nevertheless requires external scrutiny and a more comprehensive bibliographic check.

The next quantity is $R_2(n, k)$, the maximum of $|\tau\langle a, b \rangle|$ over transition pairs and output maps of minimal n -state DFAOs using k labels, as in section 10. We have proved universal upper bounds, not a formula for $R_2(n, k)$, and outside the finite range of theorem 10.1 this article does not establish the optimality of previously proposed lower-bound families or settle the remaining open questions in [1, Section 5].

The chromatic estimate suggests a concrete refinement program in three steps: optimize over collision pairs, count proper colorings for their unions, and analyze when all improper colorings are in fact reachable. The first two steps produce obstructions; the third is a constructive saturation problem of a different nature. Confusing them would turn a valid upper bound into an unjustified equality claim.

A different question is whether the uniform additive deficit can be strengthened in n as well as in k . At fixed k the quantity $k! - g(k)$ is independent of n , whereas a collision-orbit graph can exclude a number of colorings growing exponentially in n ; isolated vertices make that growth particularly transparent, since each one multiplies the proper-coloring count by k . Extracting the best universal bound from all possible cycle structures is a separate task from proving nonattainment, and is what [section 9](#) begins.

Finally, a proof-assistant formalization would be valuable. The argument uses only finite functions, elementary graph coloring, and elementary group actions, and its main logical steps are already isolated as explicit lemmas.

12.4 Group structure beyond cyclicity

The subgroup-image statement in [theorem 8.1](#) separates two independent requirements for full reachability: enough proper colorings must exist, and the permutation group must induce enough output symmetries on them. For an abelian permutation group, the second requirement already fails at three outputs. For a general group, one can ask which output symmetric groups occur as quotients of subgroups and how that restriction interacts with the chromatic number of its collision-orbit graphs.

This viewpoint is not limited to the numerical gap $k! - g(k)$. It supplies an obstruction to a whole family of colorings being in one orbit and identifies why noncommuting input permutations are effective in the ternary witness.

12.5 Conclusion

The central result is the uniform inequality

$$|\tau\langle a, b \rangle| \leq k^n - k! + g(k) < k^n \quad (3 \leq k \leq n).$$

Its proof has a concrete mechanism. A singular letter collapses a pair; the permutation letter moves that pair around a 3-colorable graph; proper colorings can then be reached only by permutation powers. Output relabeling creates symmetries that a cyclic orbit cannot realize. This gives both a quantitative proposed resolution of the selected problem and a polynomial-time procedure for exhibiting a missing reversal state.

Counting the full graph-coloring obstruction, rather than one relabeling orbit, gives the stronger structural bound. That bound establishes the reported finite optimality results when combined with the published lower construction. The unrestricted exact binary maximum remains a separate question beyond what is proved here.

A Explicit small witnesses

The following arrays use state labels $0, \dots, n-1$. Except for the $(5, 4)$ witness, the permutation is the full cycle $p = (1, 2, \dots, n-1, 0)$. The singular arrays used are

n	s
3	(0, 0, 2)
4	(0, 0, 3, 2)
5	(0, 0, 3, 4, 2)
6	(0, 3, 0, 4, 5, 1).

Suitable output arrays are

n	k	τ	$ \tau\langle p, s \rangle $
3	3	(0, 1, 2)	24
4	3	(0, 1, 0, 2)	67
4	4	(0, 1, 2, 3)	176
5	3	(0, 1, 0, 1, 2)	218
5	5	(0, 1, 2, 3, 4)	2110
6	3	(0, 0, 1, 1, 2, 2)	699
6	4	(0, 0, 1, 1, 2, 3)	3526
6	5	(0, 0, 1, 2, 3, 4)	12031
6	6	(0, 1, 2, 3, 4, 5)	32262.

For $(n, k) = (5, 4)$ use instead

$$p = (1, 0, 3, 4, 2), \quad s = (2, 1, 2, 3, 0), \quad \tau = (0, 1, 2, 3, 3),$$

whose orbit has size 826. Every listed machine is accessible from state 0 and is output-minimal; both properties are checked in the supplied code. The arrays are explicit witnesses only. Their use does not assert that every machine with the same collision graph has the same orbit size.

B Evaluating the finite bounds

The Stirling numbers needed in (9.5) are computed by the recurrence

$$\begin{Bmatrix} u \\ i \end{Bmatrix} = i \begin{Bmatrix} u-1 \\ i \end{Bmatrix} + \begin{Bmatrix} u-1 \\ i-1 \end{Bmatrix},$$

with $\begin{Bmatrix} 0 \\ 0 \end{Bmatrix} = 1$ and all impossible indices zero. This follows by considering whether the last element joins one of i existing blocks or forms a new singleton block. Thus no symbolic algebra package is needed.

The possible permutation orders can be enumerated recursively through nondecreasing cycle lengths. If the remaining number of points is r and all remaining parts must be at least j , choose a first part $i \geq j$ and combine i by least common multiple with an order from a partition of $r - i$ having parts at least i . Retaining only distinct resulting orders substantially reduces repeated work. The base case for zero remaining points is the singleton set $\{1\}$.

A direct evaluator then performs the following operations:

```
orders[r] = orders obtained from every partition of r
H(r, t)   = max(lcm(t, o) for o in orders[r])

candidates = [nonsurjective_gap, permutation_gap,
              two_singular_gap]

for m in 2..n:
    for each proper positive divisor d of m:
        add C_((m/d)(k)^d * k^(n-m) - H(n-m, m))

for a in 1..n-1:
    for b in a..n-a:
        d = gcd(a, b)
        r = n-a-b
        period = H(r, lcm(a,b))
        if k == 3 and d == 1:
```

```

    period = max(H(r,a), H(r,b))
    add B_k(a/d,b/d)^d * k^r - period

upper = k^n - min(candidates)

```

The lower evaluator independently scans coprime splits $1 < \ell < m$ with $\ell + m = n$ and maximizes (10.1). All arithmetic is exact. The data file retains both results rather than storing only a Boolean equality flag. It therefore records the actual extremal values, including numbers far larger than those feasible for literal coloring-orbit enumeration.

C A complete three-state reachability certificate

For the maps of (9.8), a displayed word $x_1 \cdots x_r$ denotes successive pullbacks, so the resulting coloring is $\tau \circ x_1 \circ \cdots \circ x_r$. The empty word is ε . Each row can be checked by direct tuple substitution; no search program is needed to verify the certificate once this table is given. This is the one place in the article where an exact orbit size is established entirely by hand.

Coloring	Word	Coloring	Word
(0, 0, 0)	<i>bab</i>	(1, 1, 2)	<i>aabaab</i>
(0, 0, 1)	<i>abaab</i>	(1, 2, 0)	<i>a</i>
(0, 0, 2)	<i>b</i>	(1, 2, 1)	<i>aabaaba</i>
(0, 1, 0)	<i>abaaba</i>	(1, 2, 2)	<i>aabaa</i>
(0, 1, 1)	<i>abaa</i>	(2, 0, 0)	<i>baa</i>
(0, 1, 2)	ε	(2, 0, 1)	<i>aa</i>
(0, 2, 0)	<i>ba</i>	(2, 0, 2)	<i>baaba</i>
(0, 2, 2)	<i>baabaa</i>	(2, 1, 1)	<i>aabaabaa</i>
(1, 0, 0)	<i>abaabaa</i>	(2, 1, 2)	<i>aaba</i>
(1, 0, 1)	<i>aba</i>	(2, 2, 0)	<i>baab</i>
(1, 1, 0)	<i>ab</i>	(2, 2, 1)	<i>aab</i>
(1, 1, 1)	<i>abab</i>	(2, 2, 2)	<i>aabab</i>

There are 24 distinct colorings in the table. Combined with the triangle obstruction excluding the other three, this proves the exact orbit size without relying on a claim that a program has examined every word. The same table is stored in `results/sharp_n3_orbit.json` as 24 pairs of a coloring and a word.

D Proof audit: points where the argument could go wrong

The main proof has a short dependency chain:

collision pair $\implies$ three-colorable orbital graph $\implies$ surjective proper k -coloring,
 fiber-partition stabilizer in a cyclic group $\implies |\text{Orb}_\tau \cap \mathcal{R}(c)| \leq g(k)$,
 rightmost singular factor $\implies$ proper reachable states use only the permutation,
 these three statements $\implies k! - g(k)$ missing states.

The cases with two singular or two invertible generators are handled separately and do not require the cyclic graph argument. The following points are stated explicitly because omitting any one of them can make an attractive informal argument incomplete or incorrect. This list merges two separately written

audit lists, and every item of each is retained: the two overlap on the rightmost-factor and surjectivity points, but each carried items the other lacked, so a shorter list would silently drop a genuine warning.

The side of composition. A word containing a singular map must be factored as $u \circ s \circ p^j$, using its *rightmost* singular factor. The collapsed input pair is then $p^{-j}\{x, y\}$. Factoring on the other side without adjusting the argument would not identify a fixed graph edge.

Surjectivity in the relabeling argument. From $\sigma \circ c = \rho \circ c$ one can conclude $\sigma = \rho$ only on the image of c . The construction therefore refines a proper coloring until it uses every output label before invoking the symmetric group; this is also what makes the $k!$ relabelings of that coloring pairwise distinct, so that the counting argument is not silently overcounting.

The fiber partition is stabilized setwise, not blockwise. In the formulation of theorem 5.2, H consists of the group elements that permute the fibers of c among themselves. Requiring each block to be preserved individually would give a smaller subgroup, a trivial induced color permutation, and a bound that proves nothing. Conversely, allowing arbitrary set maps rather than elements of the acting group would give no subgroup at all.

A cyclic component of size two is a single edge. When the quotient length is 2, the component of the collision-orbit graph consists of one undirected edge, not a two-edge multigraph on the same pair. The chromatic factor is $k(k-1)$, which is what (9.3) returns at $L=2$; treating it as a genuine 2-cycle with two parallel edges would be a modeling error even though the count happens to agree.

Different actions commute; arbitrary output permutations do not. Equation (2.1) is associativity of postcomposition and precomposition. The contradiction is that putting two noncommuting output permutations in one cyclic input orbit would force them to commute on a surjective coloring. No commutativity of all function composition is assumed.

Distinct colorings versus distinct words. The orbit size counts functions $Q \rightarrow \Delta$, not products or input words. The subgroup-image proof and all BFS implementations deduplicate colorings explicitly.

Proper colorings need not be surjective in the counting bound. The chromatic polynomial counts all proper maps into a fixed k -symbol alphabet. This is exactly the set excluded by singular-containing words. Surjectivity is needed for the factorial relabeling argument, not for (9.1).

Inaccessible proper colorings do not make improper ones accessible. The argument shows only that proper colorings are unreachable through words containing the singular letter. It says nothing about which improper colorings occur, and in particular does not license the converse reading that everything not excluded by the graph is reached. That converse is the saturation property discussed in section 9, and it has to be established separately for any claimed equality.

A graph upper bound does not create a witness. Knowing that at most $P_G(k) - T$ colorings are lost gives only an upper bound on orbit size. Lower witnesses and their reachability must be supplied separately. The finite optimality statements explicitly combine a universal upper calculation with a lower construction or a complete orbit search of an explicit machine.

References

- [1] Sylvie Davies. *State Complexity of Reversals of Deterministic Finite Automata with Output*. arXiv:1705.07150v2, 17 October 2017. <https://arxiv.org/abs/1705.07150>. Version-specific text: <https://arxiv.org/html/1705.07150v2>. The selected open problem is Section 5, Problem 1, on printed page 17; the orbit characterization is Proposition 4, ternary attainability is Theorem 1 and Corollary 2, the small maxima used as regression checks are Table 3, and the lower construction used in the finite comparison is Corollary 3. A conference version is associated with CIAA 2018, doi:10.1007/978-3-319-94812-6_12; all numbering and page references here are to the arXiv version actually consulted.

- [2] Markus Holzer and Barbara König. *On deterministic finite automata and syntactic monoid size*. Theoretical Computer Science **327**(3), 319–347, 2004. doi:10.1016/j.tcs.2004.04.010. Background on the two-generated transformation-monoid problem; not required as a black-box theorem for the principal obstruction proof. Theorem 13 contains the classical bound $n^n - n! + g(n)$, to which the present main bound specializes at $k = n$; that special case is not claimed as new. Sections 4–5 use graph colorings and chromatic polynomials in the transformation-monoid setting, so graph coloring as a method is likewise not claimed here as an invention.
- [3] Marc Deléglise and Jean-Louis Nicolas. *The Landau Function and the Riemann Hypothesis*. arXiv:1907.07664, 2019. <https://arxiv.org/abs/1907.07664>. Cited only for the standard name and definition of $g(k)$; no analytic result from this work is used.